

A scalar-valued counterexample to Rao's converse question

arXiv:math/0312116, Corollary 14

Source Question

Rao proves that every very smooth point T of $\mathcal{K}(X, Y)$ attains its norm at a unique unit vector, up to scalar multiples, and that the image vector is a very smooth point of Y . Immediately after Corollary 14, the paper asks whether the converse is always true [1].

COROLLARY 14.

Every very smooth point T of $K(X, Y)$ attains its norm at a unique unit vector (up to constants) x_0 and $T(x_0)$ is a very smooth point of Y .

Proof. Let T be a very smooth point of $K(X, Y)$. As noted before, there exists a $x_0^{**} \in \partial_e X_1^{**}$ and a $y_0^* \in \partial_e Y_1^*$ such that $x_0^{**} \otimes y_0^*$ is the unique element of $\partial_e K(X, Y)_1^*$ that attains its norm at T . Since T is very smooth, $x_0^{**} \otimes y_0^*$ is a point of $w^* - w$ continuity. Therefore, $x_0^{**} = x_0 \in X$. Hence $1 = y_0^*(T(x_0)) \leq \|T(x_0)\| \leq \|T\| = 1$. Thus $\|T\| = 1 = \|T(x_0)\|$. That $T(x_0)$ is very smooth again follows from the above theorem. $\square$

We do not know if the converse of the above theorem is always true. However when $Y = C(K)$, for a compact set K we have the following complete description of very smooth points of $\mathcal{K}(X, C(K))$. It is well known that $\mathcal{K}(X, C(K))$ can be identified with the space of vector-valued functions $C(K, X^*)$. Since in this case the arguments given during the proof of Theorem 12 are much simpler we present the complete proof.

Figure 1: Corollary 14 and the converse question on page 58 of Rao's paper.

Proof Intuition

The necessary condition in Corollary 14 only asks where the operator attains its norm on the original unit sphere. For a functional on ℓ^1 , this detects only coordinates where the coefficient sequence actually reaches its supremum. Smoothness of the same functional as a point of $\ell^\infty = (\ell^1)^*$ is more sensitive: if the coefficients approach their supremum along the tail, then free-ultrafilter limits also norm the sequence. Thus one can force unique norm-attainment on ℓ^1 while keeping several norming functionals in $(\ell^\infty)^*$.

Counterexample

Theorem 1. *The converse question after Corollary 14 in [1] has a negative answer. There are Banach spaces X, Y and a compact operator $T \in \mathcal{K}(X, Y)$ such that T attains its norm at a unique unit vector up to scalar multiples and $T(x_0)$ is a very smooth point of Y , but T is not a very smooth point of $\mathcal{K}(X, Y)$.*

Proof. Let $X = \ell^1$, let $Y = \mathbb{K}$ be the scalar field, and define $T : \ell^1 \rightarrow \mathbb{K}$ by

$$Tx = \sum_{n=1}^{\infty} a_n x_n, \quad a_1 = 1, \quad a_n = 1 - \frac{1}{n} \quad (n \geq 2).$$

Since Y is one-dimensional, T is compact. Also $\|T\| = \|(a_n)\|_\infty = 1$.

We first check the hypothesis suggested by the converse. If $\|x\|_1 = 1$ and $|Tx| = 1$, then

$$1 = |Tx| \leq \sum_{n=1}^{\infty} a_n |x_n| \leq \sum_{n=1}^{\infty} |x_n| = 1.$$

For every $n \geq 2$, $a_n < 1$. Equality in the second inequality therefore forces $x_n = 0$ for all $n \geq 2$. Hence $x = \lambda e_1$ with $|\lambda| = 1$. Conversely, all such scalar multiples attain the norm. Thus T attains its norm at the unique unit vector e_1 , up to constants. Moreover $T(e_1) = 1$, and every unit vector in the one-dimensional space $\mathbb{K}$ is very smooth.

It remains to show that T is not very smooth as a point of $\mathcal{K}(\ell^1, \mathbb{K})$. The space $\mathcal{K}(\ell^1, \mathbb{K})$ is just $\mathcal{L}(\ell^1, \mathbb{K}) = (\ell^1)^* = \ell^\infty$ isometrically, and T corresponds to $a = (a_n) \in \ell^\infty$.

Let $\delta_1 \in (\ell^\infty)^*$ be first-coordinate evaluation. Then $\|\delta_1\| = 1$ and $\delta_1(a) = 1$. Let $\mathcal{U}$ be a free ultrafilter on $\mathbb{N}$, and define

$$L_{\mathcal{U}}(b) = \lim_{\mathcal{U}} b_n \quad (b = (b_n) \in \ell^\infty).$$

This is a norm-one functional on ℓ^∞ . Since $a_n \rightarrow 1$, we have $L_{\mathcal{U}}(a) = 1$. The two norming functionals are distinct, for example $\delta_1(e_1) = 1$ while $L_{\mathcal{U}}(e_1) = 0$.

Therefore a is not a smooth point of ℓ^∞ . A very smooth point of a Banach space is, in particular, a smooth point: indeed, two distinct norm-one functionals on the space that norm the point give two distinct norm-one functionals on the bidual via the canonical embedding. Hence T is not a very smooth point of $\mathcal{K}(\ell^1, \mathbb{K})$. $\square$

Verification and Scope

The example satisfies the exact hypotheses proposed by the converse: compactness is automatic for scalar-valued operators, norm-attainment is unique up to scalars, and the image vector lies in a one-dimensional very smooth range. The conclusion fails because the operator is not even a smooth point of $\mathcal{K}(\ell^1, \mathbb{K})$.

No computational verification is needed.

Novelty Check

The run indexes were searched for 0312116, **very smooth**, **unique norm**, **norm-attain**, **Rao**, **operator converse**, and **compact operator**; no prior packet or attempt was found. The local parsed source corpus was searched for the exact question and adjacent key phrases. Web searches for the exact converse sentence, for the paper title together with **converse**, and for phrases involving $\mathcal{K}(X, Y)$ and unique norm-attainment found the arXiv source page but no separate erratum or known answer. The novelty confidence is therefore local-and-bounded rather than exhaustive.

References

- [1] T. S. S. R. K. Rao, *Very smooth points of spaces of operators*, arXiv:math/0312116, 2003. <https://arxiv.org/abs/math/0312116>

Human Review: Very Smooth Operator Converse Counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Verified.

Verdict: The proof is correct. The example does show that the converse question after Corollary 14 in Rao's paper is not true in general.

Human Comments

The proposed counterexample is elementary and convincing. Let

$$X = \ell_1, \quad Y = \mathbb{K},$$

and let $T \in \mathcal{K}(\ell_1, \mathbb{K})$ be the scalar-valued operator represented by the sequence

$$a = (a_n) \in \ell_\infty, \quad a_1 = 1, \quad a_n = 1 - \frac{1}{n} \quad (n \geq 2).$$

Since the range is one-dimensional, T is compact. Also

$$\|T\| = \|a\|_\infty = 1.$$

The operator T attains its norm only at e_1 , up to scalar multiples. Indeed, if $\|x\|_1 = 1$ and $|Tx| = 1$, then

$$1 = |Tx| \leq \sum_{n=1}^{\infty} a_n |x_n| \leq \sum_{n=1}^{\infty} |x_n| = 1.$$

Since $a_n < 1$ for every $n \geq 2$, equality forces $x_n = 0$ for all $n \geq 2$. Hence $x = \lambda e_1$, with $|\lambda| = 1$. Thus the norm-attaining vector is unique up to scalars.

Moreover,

$$T(e_1) = 1 \in \mathbb{K},$$

and every unit vector in the one-dimensional space $\mathbb{K}$ is very smooth. Therefore the hypotheses suggested by Rao's converse question are satisfied.

However, T is not a very smooth point of $\mathcal{K}(\ell_1, \mathbb{K})$. In fact,

$$\mathcal{K}(\ell_1, \mathbb{K}) = \mathcal{L}(\ell_1, \mathbb{K}) = (\ell_1)^* = \ell_\infty$$

isometrically, and T corresponds to the sequence a . This sequence is not a smooth point of ℓ_∞ . The coordinate functional

$$\delta_1(b) = b_1$$

has norm one and satisfies

$$\delta_1(a) = 1.$$

On the other hand, if $\mathcal{U}$ is a free ultrafilter on $\mathbb{N}$, then

$$L_{\mathcal{U}}(b) = \lim_{\mathcal{U}} b_n$$

defines a norm-one functional on ℓ_∞ , and since $a_n \rightarrow 1$, one has

$$L_{\mathcal{U}}(a) = 1.$$

The two norming functionals are distinct; for instance,

$$\delta_1(e_1) = 1, \quad L_{\mathcal{U}}(e_1) = 0.$$

Thus a is not a smooth point of ℓ_∞ . Consequently T is not even smooth as a point of $\mathcal{K}(\ell_1, \mathbb{K})$, and therefore it cannot be very smooth.

Short Version

The example satisfies the proposed converse hypotheses: T is compact, it attains its norm at a unique unit vector up to scalars, and $T(e_1)$ is very smooth in the scalar range. Nevertheless, T corresponds to a nonsmooth point of ℓ_∞ , because the sequence

$$(1, 1/2, 2/3, 3/4, \dots)$$

is normed both by first-coordinate evaluation and by free-ultrafilter limits. Hence T is not very smooth.

Review Outcome

This should be classified as a correct counterexample. It proves that the converse question after Corollary 14 is false in general.